

Thuc Nguyen Phong Pham

University of British Columbia | 3rd Year

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GitHub: github.com/prcpham-dev **Portfolio:** prcpham.com

Experience

Software Engineering Intern – Frontend Focus

Apr 2023 – Aug 2023

Tinh Van Software

- Developed responsive web components using React.js, Node.js, TypeScript, Bootstrap, and Tailwind CSS for a client-facing platform.
- Collaborated with backend engineers to integrate and test RESTful APIs using Postman.
- Participated in deployment and planning using Git.
- Gained hands-on exposure to backend infrastructure involving FastAPI and Uvicorn.

Software Engineering Intern – Full-Stack

Apr 2024 – Aug 2024

Tinh Van Software

- Designed and developed a Retrieval-Augmented Generation (RAG) web app with PDF uploader and contextual AI chat capabilities.
- Implemented backend in Python using FastAPI, LangChain, and use Google Gemini to support semantic search and response generation.
- Integrated Redis and Celery for asynchronous task queues, improving responsiveness and scalability.
- Built a responsive frontend using React.js and Bootstrap.
- Deployed on a cloud VPS using Docker, Nginx, HTTPS (SSL), and environment-managed API keys.
- Created RESTful API endpoints for document processing and chat flow, supporting up to 3MB PDF uploads.

Education

University of British Columbia

Expected May 2026

Bachelor of Science in Combined major science

Relevant Coursework: Data Structures, Algorithms, Operating Systems, Databases, Machine Learning

Projects

ChatDocument

FastAPI, LangChain, Redis, Celery, React.js, Bootstrap, Docker, Nginx

- Developed a Retrieval-Augmented Generation (RAG) app enabling users to upload and analyze PDFs with contextual AI chat.
- Integrated Google Gemini (2M tokens) with LangChain for embeddings, semantic search, and long-context reasoning.
- Built a scalable backend with FastAPI, offloading heavy tasks asynchronously using Redis and Celery.
- Designed a responsive frontend in React.js and Bootstrap for file upload, chat interaction, and insights display.
- Deployed on a VPS with Docker, Nginx, and SSL certificates, securing API keys via environment management.

InstaSave

Chrome Extension, JavaScript, MutationObserver

- Created a browser extension that injects a “Save” button into Instagram posts and stories for quick media downloads.
- Used DOM manipulation and MutationObserver to dynamically detect and modify Instagram’s interface.
- Delivered lightweight functionality to open or save images and videos directly from the browser.

Shufa Character Downloader

Selenium, asyncio, Tkinter, ChromeDriver

- Built a desktop application with a Tkinter-based UI to automate downloading of Chinese calligraphy images.
- Automated scraping with Selenium WebDriver to fetch and filter results for the selected calligrapher.
- Leveraged asyncio to run multiple download tasks concurrently without blocking the UI.
- Organized images into a structured local directory and displayed progress and logs in the UI.

GhostRun

HTML5 Canvas, JavaScript, LocalStorage

- Built an endless runner game where a ghost character flips gravity to dodge pillars, inspired by Flappy Bird.
- Implemented physics with gravity inversion, collision detection, and difficulty scaling using vanilla JavaScript.
- Designed game loop with `requestAnimationFrame` for smooth rendering on HTML5 Canvas.
- Added HUD with score, high score persistence via `localStorage`, and interactive pause/restart controls.
- Deployed on GitHub Pages for browser play: prepham-dev.github.io/GhostRun.

SpeedCube

React.js, Three.js, Tailwind CSS

- Built an interactive 3D Rubik’s Cube simulator with real-time rendering in a React web app.
- Implemented a precision timer, customizable keybinds, and state control (shuffle, reset).
- Designed a modern, responsive UI with Tailwind CSS for smooth cross-device use.

YoutubeShorts_ClipMixer

OpenAI API, AssemblyAI, Tkinter, MoviePy

- Built a GUI-based tool to automate YouTube Shorts creation from user-provided or AI-generated scripts.
- Integrated APIs for narration, transcription, and captions to streamline video production.
- Assembled and edited clips with MoviePy and FFmpeg, adding background music and rendering final output.
- Provided a Tkinter interface to manage script input, API keys, and video preview/editing.

Skills

- **Languages:** JavaScript, TypeScript, Python, Java, Swift, SQL, HTML/CSS, C++
- **Frameworks/Libraries:** React.js, Node.js, SwiftUI, Tailwind CSS, Three.js, MoviePy, LangChain, FastAPI, Express.js
- **Tools/Technologies:** Docker, Git, Postman, Redis, Celery, Uvicorn, REST APIs, OpenAI API, AssemblyAI, Selenium/WebDriver, PostgreSQL, AWS